JE Agent

Journal-entry testing, **decided.**

Project Report

JE Agent — Journal Entry Testing AI Agent

ISA 240 / AS 2401 · V1.1

PREPARED BY

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Journal Entry Testing, Decided.

JE Agent — an AI-assisted audit assistant for journal entries (ISA 240 / AS 2401)

This report explains **what JE Agent is, how it thinks, and what the evidence says about how well it works** — without requiring a technical background. Deep technical detail lives in the repository (`DESIGN.md`, `PROJECT_REPORT.md`); this document focuses on the logic and the proof.

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1 · The Problem

Auditors are required to test journal entries for fraud risk (ISA 240 / AS 2401). In practice this means reading through thousands — sometimes hundreds of thousands — of accounting lines, looking for the small number of entries that deserve suspicion:

- amounts posted at round numbers,
- invoices split into pieces to stay under approval limits,
- entries posted at unusual times (period end),
- unusual user activity, unusual account combinations,
- entries that don't balance, reversals, date inconsistencies...

Doing this by hand is slow and unreliable. Doing it with a plain script produces thousands of "hits" that are almost all harmless. And doing it with AI alone raises the question every auditor must ask: *can I trust a black box?*

JE Agent's answer: don't choose between rules, AI, and humans — use all three, each where it is strongest.

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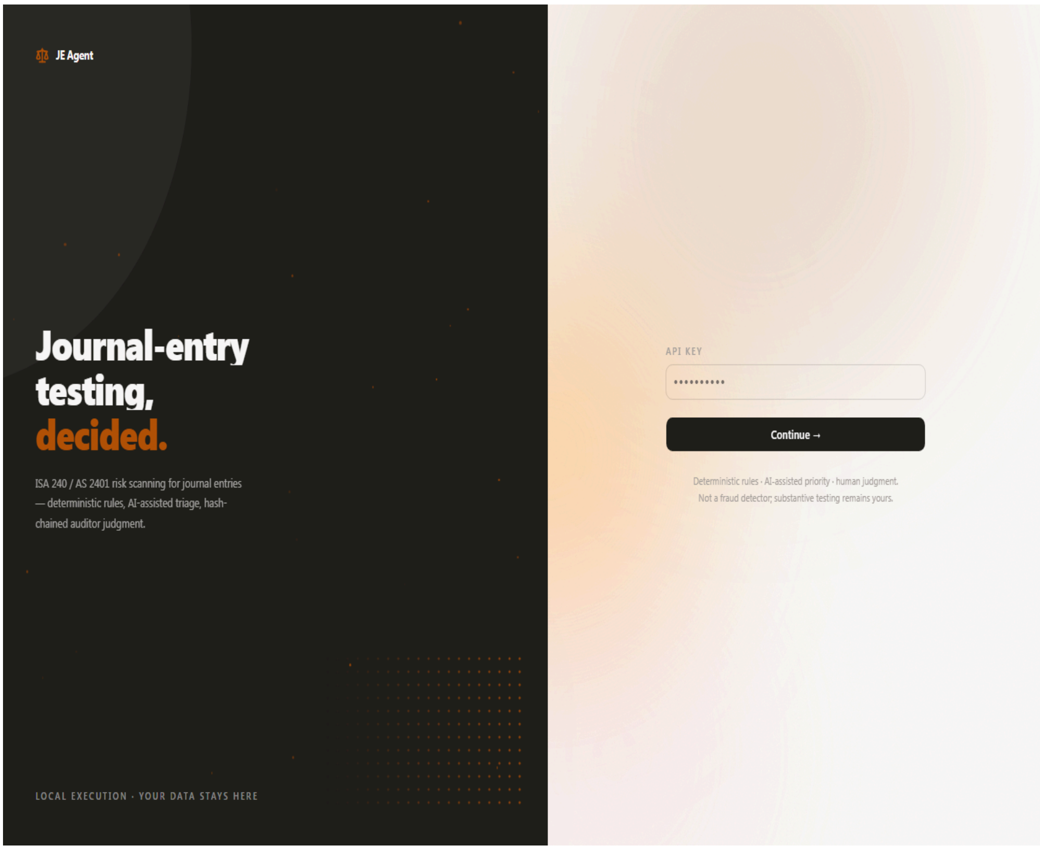
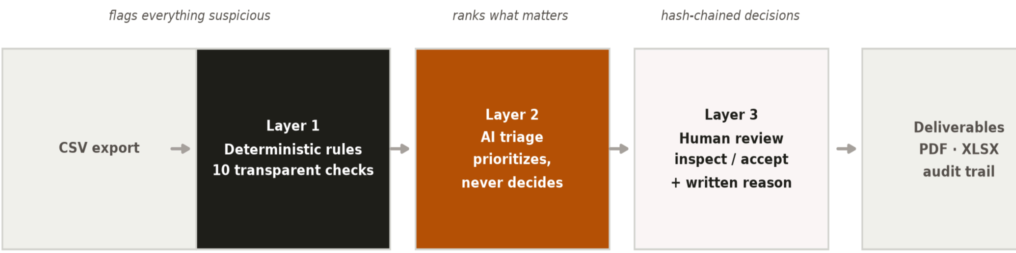
2 · The Logic — Three Layers



JE Agent processes an accounting export through three layers, arranged exactly by trustworthiness:



Layer 1 — Deterministic rules (never wrong about *what it measures*). Ten transparent rules scan every entry: round amounts, splitting, period-end timing, balance checks, rare users, odd account pairs, reversals, date divergence, high-risk users, and manual-entry concentration. Every flag comes with its reason. These rules are fast and explainable — but as we will show in §4, they also over-flag: they are a *net*, not a verdict.

Layer 2 — AI triage (prioritization, never decision). An AI model reads the flagged entries and ranks which ones look most worth a human's attention, writing a short rationale for each. Crucially, the AI **cannot finalize anything**: its output is advisory, recorded as such in the audit trail.

Layer 3 — Human judgment (where precision is made). Every entry that reaches review requires an explicit auditor decision — *inspect* or *accept* — with a mandatory written reason. Each decision is **hash-chained** to the previous one, so any later tampering would be mathematically detectable.

The pipeline then produces the deliverables an engagement file needs: a professional PDF report (in English or French), an Excel workpaper, and a list of flagged entries.

> ***Design principle:*** *the machine proposes, the human disposes. Nothing reaches "final" without a named person's recorded judgment.*

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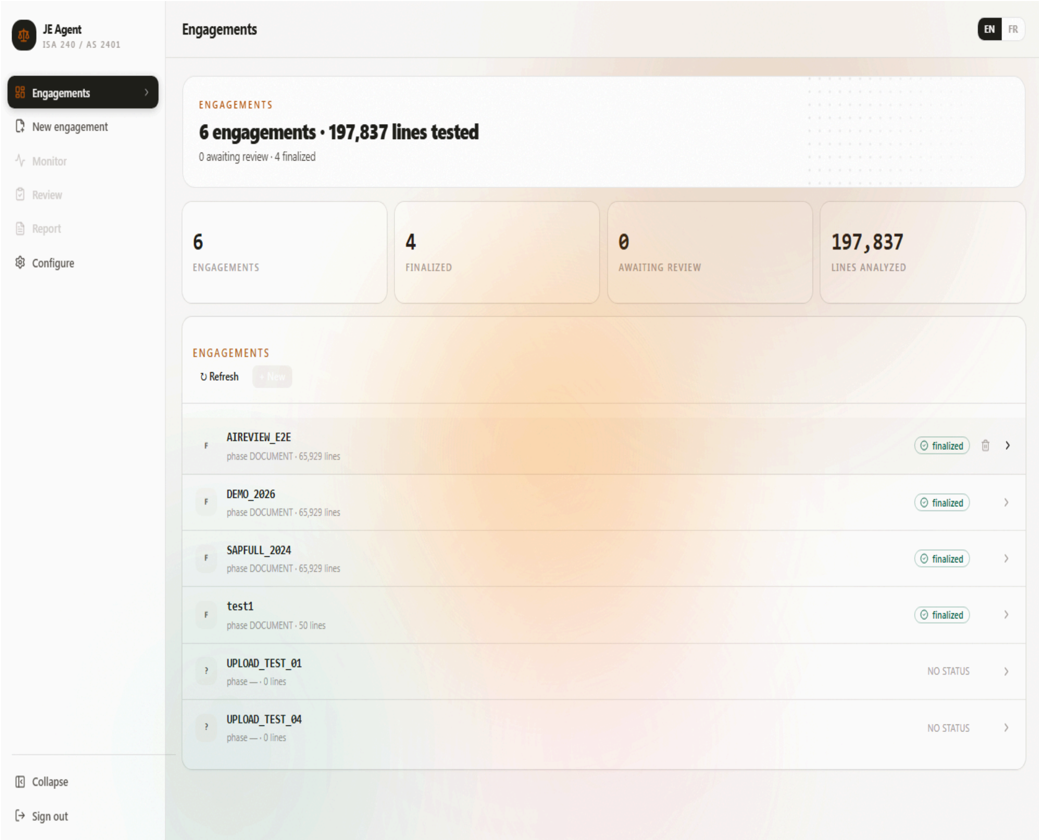
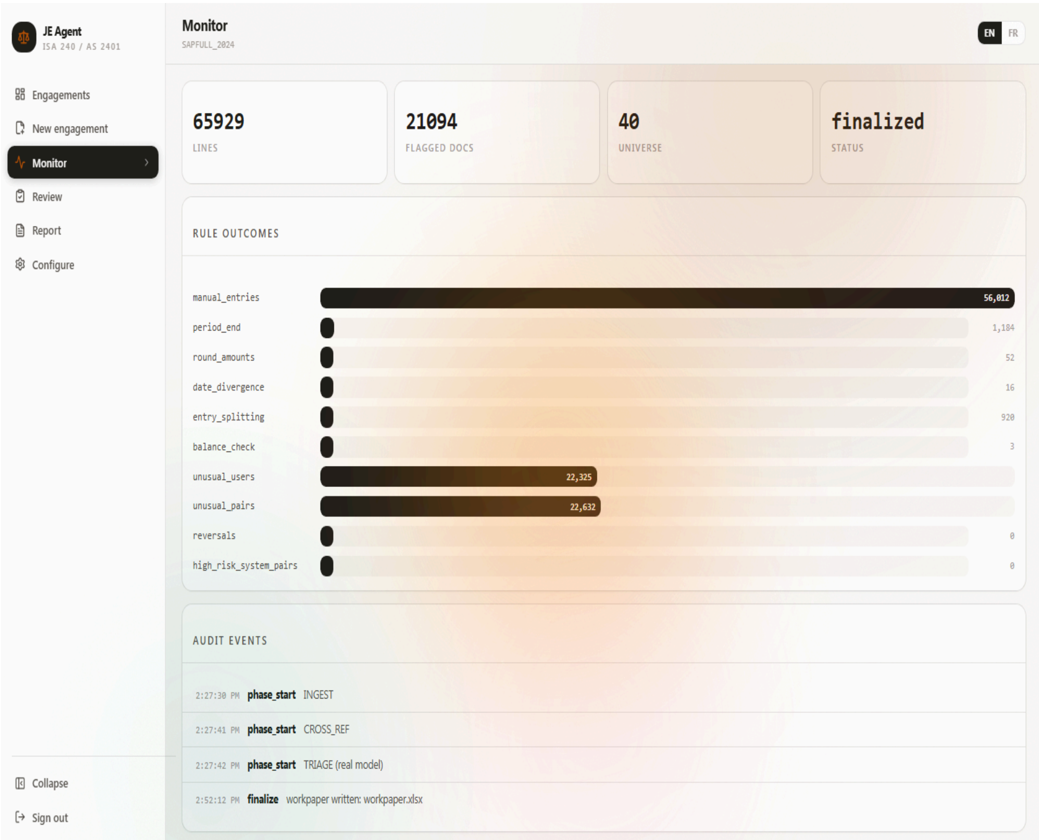
3 · The Console

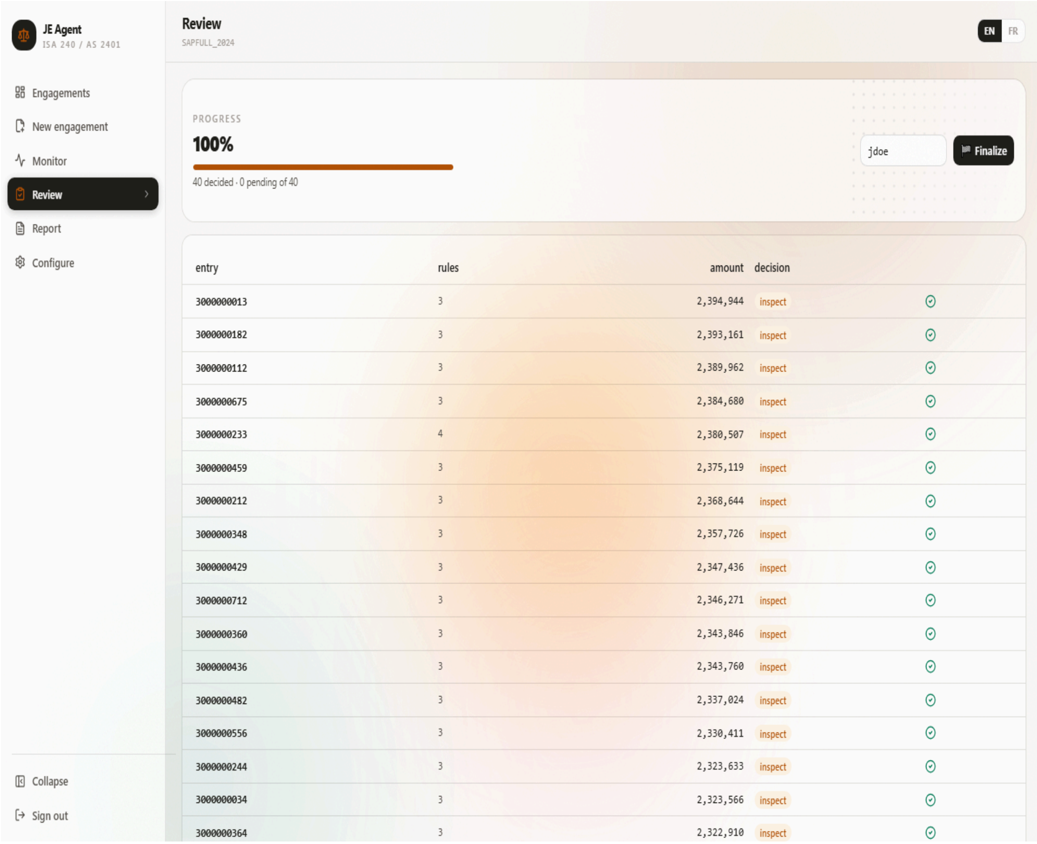
JE Agent runs locally — data never leaves the machine except optional AI triage calls (which run in zero-retention mode).



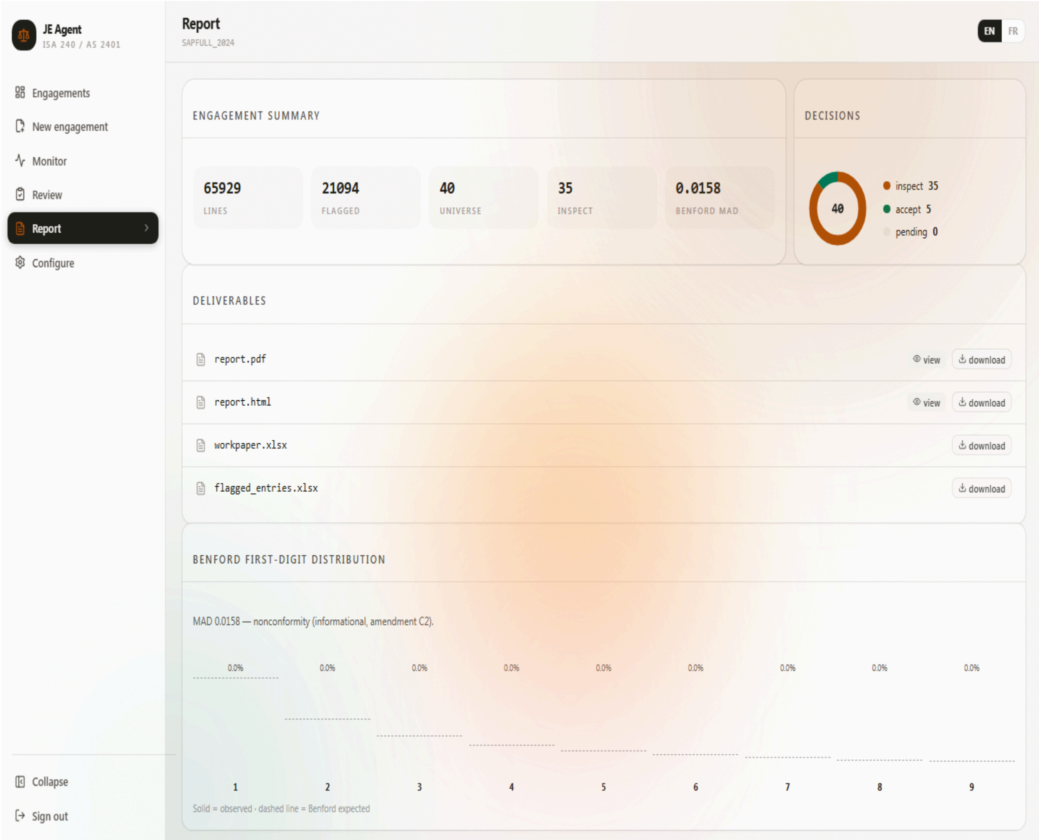
The dashboard summarizes every engagement: population sizes, finalization status, and one-click access to any run.



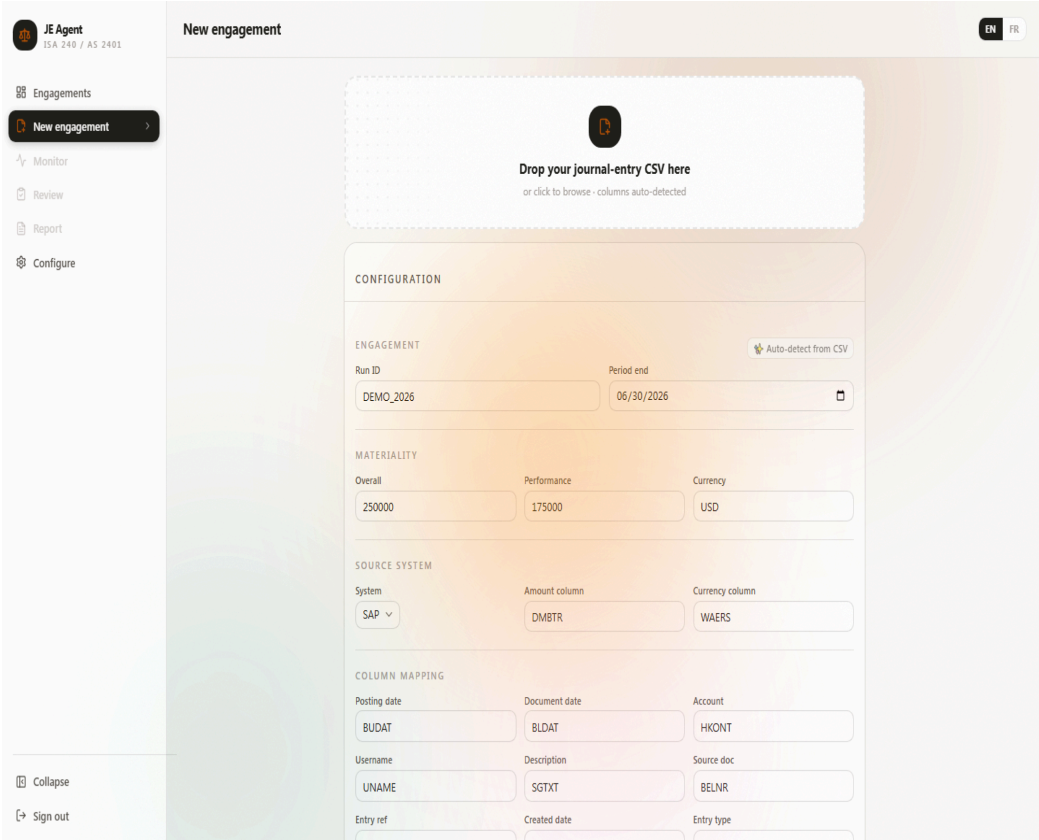
The Monitor page shows what each rule found and the full audit event timeline of the run.



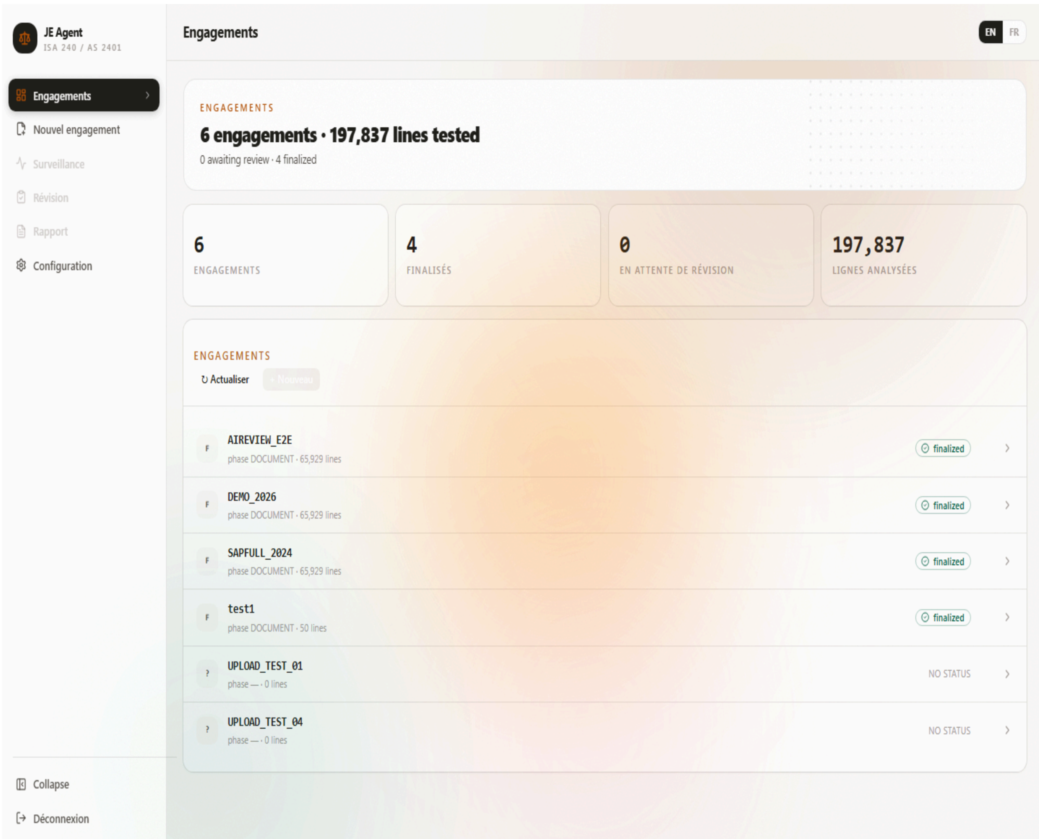
The Review queue presents entries like a ledger. Clicking an entry opens its full journal lines and explains *which rules fired*. Decisions require a written reason and are hash-chained on confirmation.



The Report page gathers deliverables (PDF, HTML, Excel workpaper), a decisions breakdown, and a Benford first-digit analysis against the theoretically expected distribution.



Creating an engagement is a single CSV upload — column mapping is auto-detected from the file itself, materiality and review mode are set in one form.



The entire interface — and the generated report — switch between English and French.

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4 · Does It Work? The Stress Test

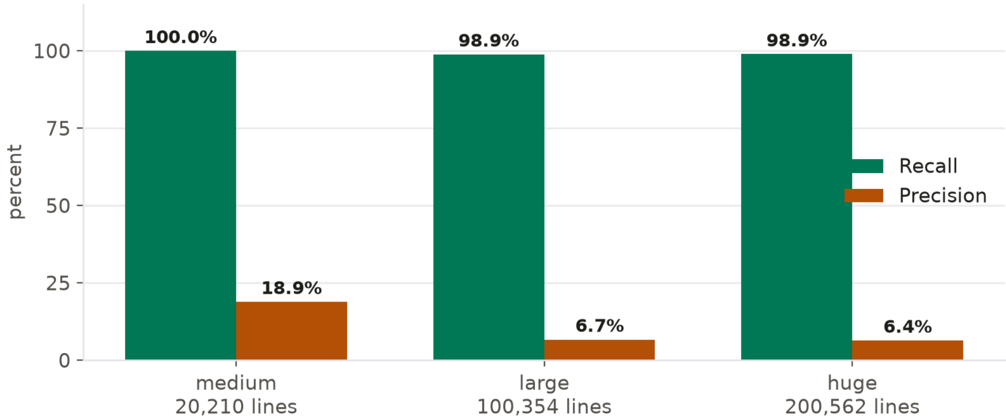
Trust claims are cheap; we measured ours. We built a **stress harness** that generates synthetic accounting populations with **known anomalies injected** — every planted anomaly is labeled with the rule that should catch it. Because the truth is known, recall and precision can be computed exactly.

Methodology

- Populations from **20,000 to 200,000+ lines**, deterministic (same seed → same data).
- Bases made deliberately **realistic, not sterile**: ~90% manual + ~10% system-posted entries, legitimately rare users, legitimately large invoices, month-start postings — noise that a naive test would hide.
- Scoring per rule: **recall** = share of planted anomalies caught; **precision** = share of flags that were real anomalies; false positives counted only against clean documents.

Headline results

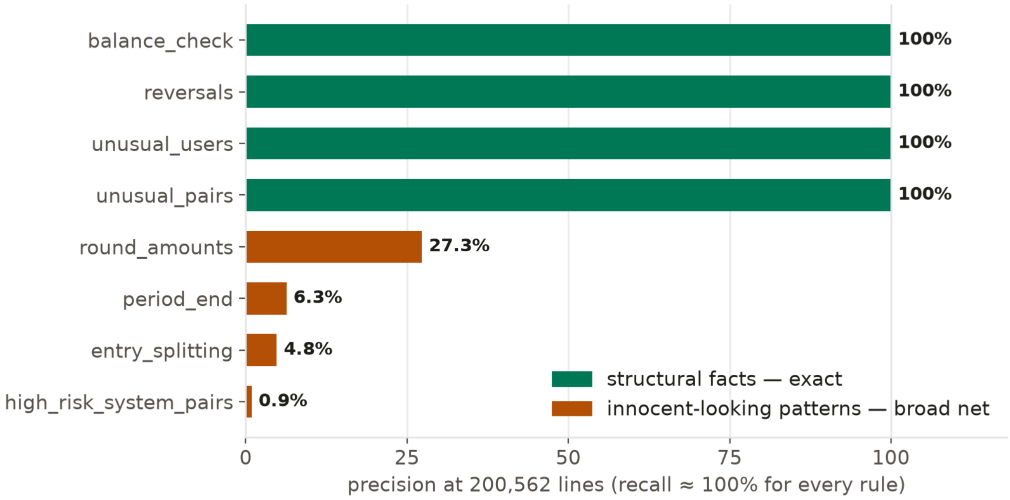
Scenario	Lines	Planted	Recall	Precision
medium	20,210	105	**100%**	18.9%
large	100,354	177	**98.9%**	6.7%
huge	200,562	281	**98.9%**	6.4%



Reading this honestly: the system *almost never misses a planted anomaly, at any scale* (200k lines ingested in ~14 seconds). But aggregate precision falls as populations grow — because benign patterns that *look* anomalous (a big round invoice, a month-start posting) grow with the population while true fraud stays rare.

Per-rule picture (200k-line scenario)

Rule	Recall	Precision
round_amounts	100%	27%
entry_splitting	100%	5%
period_end	100%	6%
unusual_pairs	96%	100%
unusual_users	100%	100%
balance_check	100%	100%
reversals	100%	100%
date_divergence	87%	100%
high_risk_system_pairs	100%	0.9%



Two clean stories emerge. Rules based on **structural facts** (balance checks, reversals, rare users, account pairs) are essentially perfect on both axes. Rules based on **patterns that have innocent explanations** (round amounts, timing, size clustering) catch everything but drown in legitimate look-alikes at scale.

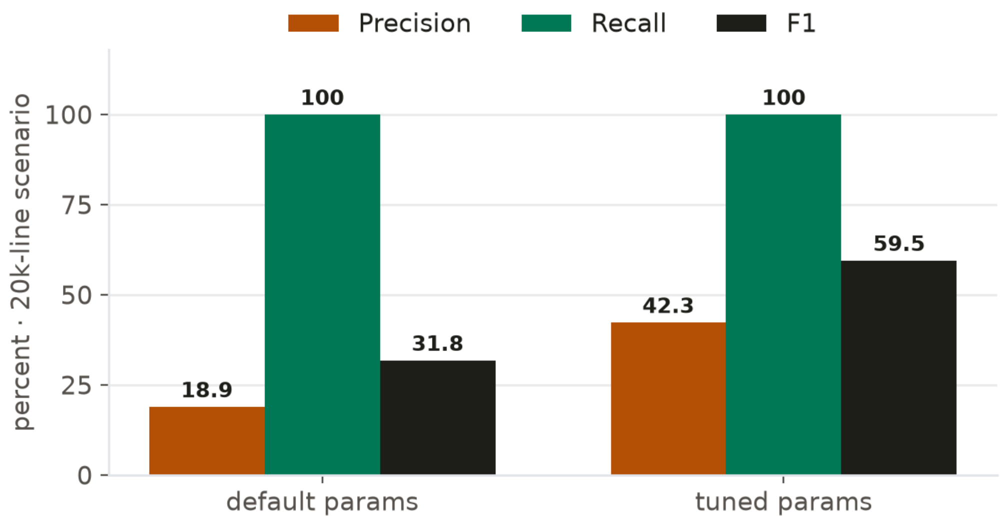
The universe check

Reviewers can only see a capped selection (the *universe*) ranked above performance materiality. In testing, **97–100% of above-materiality planted anomalies reached the reviewer** — the scoping logic does its job.

Can tuning help? Yes — measurably.

Using the same harness, two threshold changes (narrower period-end window, stricter rarity definition for system pairs) were tested A/B on the 20k scenario:

Metric	Default	Tuned
Aggregate precision	18.9%	**42.3%**
Aggregate recall	100%	**100%**



Precision more than doubled at zero recall cost — and because thresholds interact with the population, the lesson generalizes: **rule parameters must be configured against the engagement's own materiality**, and this harness makes that calibration measurable instead of guesswork.

The AI layer

On the smallest scenario, the real LLM triage leg was scored separately: it **caught 100% of planted anomalies** but recommended inspecting many clean entries too (inspect precision ≈41%). That is precisely its role in the design: a broad prioritizer feeding the human, not a decider. The three layers trade roles exactly as intended — rules never miss, AI ranks everything, the human makes precision.

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5 · Trust & Integrity

- **Hash-chained decisions** — every reviewer decision links cryptographically to the previous one; tampering is detectable.
- **Citation gates** — the AI narrative cannot be finalized unless every claim cites verifiable data from the run.
- **Deterministic core** — same input, same flags, always reproducible.

- **Human-only finalization** — four gates (review complete, procedures documented, citations valid, limitations stated) block the report until a human's decisions are in place.
- **Local-first** — engagement data stays on the machine; AI calls run in zero-retention privacy mode.

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6 · Honest Limitations

- **Precision at scale is low by construction**: the rule layer is a sieve. This is a feature of the domain (fraud is rare and mimics normality), not a bug — but users must expect to review, not read raw flag counts.
- **One config-dependent rule** (`high_risk_system_pairs`) needs the auditor to define high-risk users; unconfigured, it fires indiscriminately (0.9% precision measured).
- `date_divergence` missed ~13% of plants whose fake document dates landed inside the benign window — a known edge, documented rather than hidden.
- The synthetic populations, however realistic, are still synthetic; field behavior on live client data will differ.

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7 · Conclusion

JE Agent demonstrates that journal-entry testing can be **fast, transparent, and honest at once**: near-perfect detection of designed anomaly classes at 200k-line scale in seconds, an architecture that gives the AI no final authority, integrity guarantees that make the audit trail tamper-evident, and measured limitations published alongside the strengths. Its strongest credibility asset is the last one: **it tells you exactly what it is good at, what it is not, and shows the numbers for both.**